2.61 Rational Functions: Domain and Range

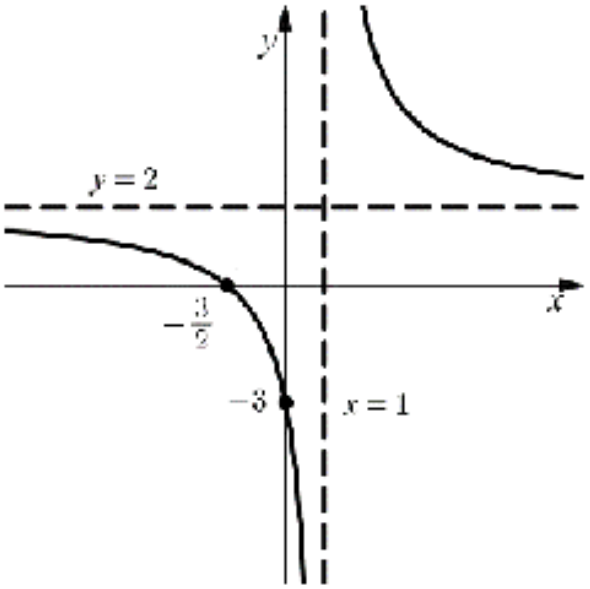
Worked Solutions

1a) The denominator is zero at $x = 1$. The domain is $\{x \in \mathbb{R} : x \neq 1\}$.

1b) $f(x) = 2 + \dfrac{5}{x-1}$. The asymptotes are $x = 1$ and $y = 2$.

1c) $f(x) = 2 + \dfrac{5}{x-1}$ can never equal 2. The range is $\{f(x) : f(x) \in \mathbb{R},\ f(x) \neq 2\}$.

1d) The graph has asymptotes $x = 1$, $y = 2$, and intercepts $(-\frac{3}{2}, 0)$ and $(0, -3)$.

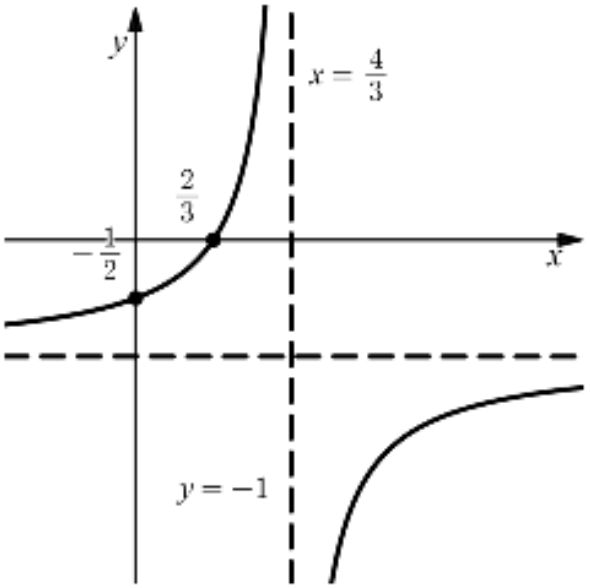


2a) The denominator is zero at $x = \frac{4}{3}$. The domain is $\{x \in \mathbb{R} : x \neq \frac{4}{3}\}$.

2b) Division gives $f(x) = -1 + \dfrac{2}{4-3x}$. The asymptotes are $x = \frac{4}{3}$ and $y = -1$.

2c) $f(x) = -1$ has no solution, so the range is $\{f(x) : f(x) \in \mathbb{R},\ f(x) \neq -1\}$.

2d) The graph has asymptotes $x = \frac{4}{3}$ and $y = -1$, and intercepts $(0, -\frac{1}{2})$ and $(\frac{2}{3}, 0)$.



3a) The denominator $x^2 + 4$ is positive for all real x, so the domain is $\mathbb{R}$.

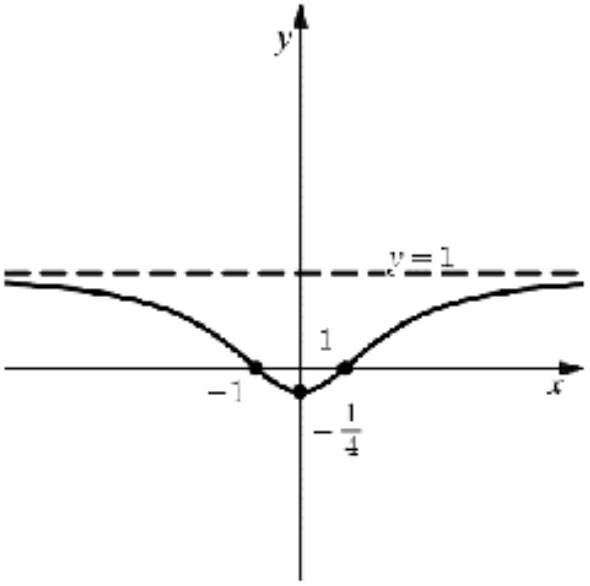
3b) The horizontal asymptote is $y = 1$.

3c) Set $y = k$:

$$k(x^2 + 4) = x^2 - 1 \quad \Longrightarrow \quad (k-1)x^2 + (4k+1) = 0.$$

For real x, $(k-1)(4k+1) \leq 0$, giving $-\frac{1}{4} \leq k \leq 1$. Since $k = 1$ is the horizontal asymptote and is not attained, the range is $\{f(x) : f(x) \in \mathbb{R}, -\frac{1}{4} \leq f(x) < 1\}$.

3d) The curve is even, has a minimum at $(0, -\frac{1}{4})$, and approaches the horizontal asymptote $y = 1$ from below.



4a) Factor the denominator: $x^2 - x - 2 = (x-2)(x+1)$. The numerator is $(3x+2)(x-1)$, so the vertical asymptotes are $x = -1, 2$; the horizontal asymptote is $y = 3$.

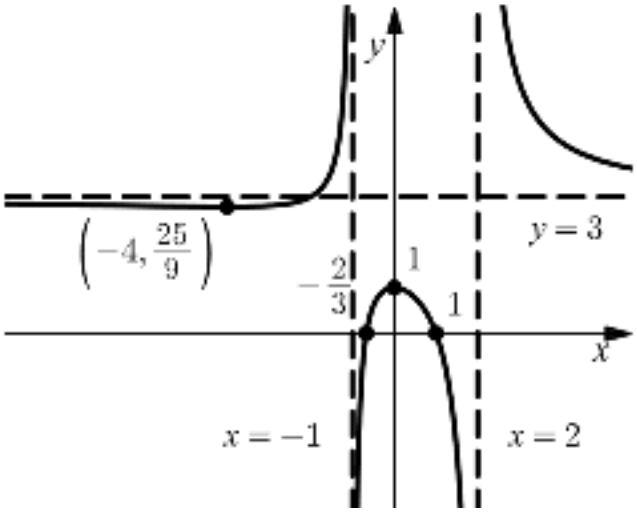
4b) Set $f(x) = k$:

$$(k-3)x^2 + (1-k)x + (2-2k) = 0.$$

For real intersections the discriminant condition gives $(k-1)(9k-25) \geq 0$, so $k \leq 1$ or $k \geq \frac{25}{9}$. Thus the range is $\{f(x) : f(x) \in \mathbb{R}, f(x) \leq 1\} \cup \{f(x) : f(x) \in \mathbb{R}, f(x) \geq \frac{25}{9}\}$.

4c) At $y = 1$, solving gives $x = 0$. At $y = \frac{25}{9}$, solving gives $x = -4$. The stationary points are $(-4, \frac{25}{9})$ and $(0, 1)$.

4d) The curve has asymptotes $x = -1$, $x = 2$, and $y = 3$; it has stationary points $(-4, \frac{25}{9})$ and $(0, 1)$.



5a) The denominator is zero at $x = -2$. The domain is $\{x \in \mathbb{R} : x \neq -2\}$.

5b) The asymptotes are $x = -2$ and $y = 1$.

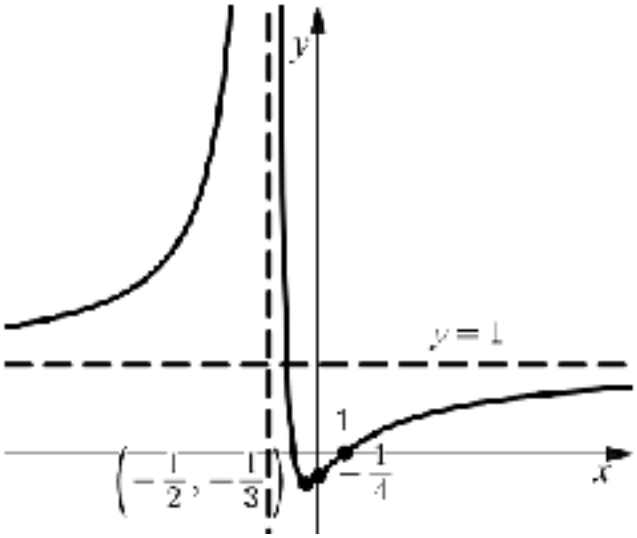
5c) Set $f(x) = k$:

$$(k-1)x^2 + 4kx + 4k + 1 = 0.$$

The discriminant condition reduces to $12k + 4 \geq 0$, so $k \geq -\frac{1}{3}$. The function has a minimum value of $-\frac{1}{3}$.

5d) The range is $\{f(x) : f(x) \in \mathbb{R}, f(x) \geq -\frac{1}{3}\}$.

5e) The graph has vertical asymptote $x = -2$, horizontal asymptote $y = 1$, and minimum $(-\frac{1}{2}, -\frac{1}{3})$.



6a) The denominator factors as $(x+5)(x+1)$. The domain is $\{x \in \mathbb{R} : x \neq -5,\ x \neq -1\}$.

6b) The vertical asymptotes are $x = -5$ and $x = -1$; since numerator and denominator have equal degree, the horizontal asymptote is $y = a$.

6c) The given horizontal asymptote is $y = 3$, so $a = 3$. Setting $f(x) = k$ gives

$$(k-3)x^2 + 6kx + 5k - 6 = 0.$$

The discriminant condition simplifies to $16k^2 + 84k - 72 \geq 0$, with equality at $k = -6$ and $k = \frac{3}{4}$. These are the y-coordinates of the stationary points.

6d) The range is $\{f(x) : f(x) \in \mathbb{R}, f(x) \leq -6\} \cup \{f(x) : f(x) \in \mathbb{R}, f(x) \geq \frac{3}{4}\}$.

7a) Since $x^2 + 1 > 0$, the domain is $\mathbb{R}$.

7b) Set $f(x) = k$:

$$(2-k)x^2 + 3 - k = 0.$$

The discriminant condition gives $2 < k \leq 3$ (the value $k = 2$ is not attained). The range is $\{f(x) : f(x) \in \mathbb{R}, 2 < f(x) \leq 3\}$.

7c) Solving $f(x) = \frac{5}{2}$ gives $x^2 = 1$, so the solution is $\{x \in \mathbb{R} : -1 \leq x \leq 1\}$.

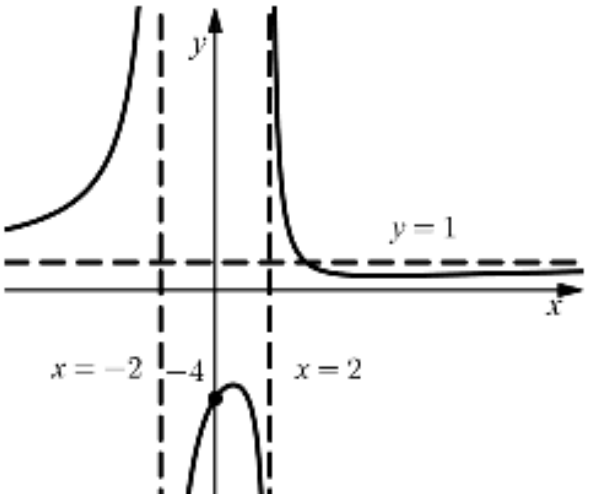
8a) The denominator is $(x-2)(x+2)$, so the domain is $\{x \in \mathbb{R} : x \neq -2,\ x \neq 2\}$.

8b) Set $f(x) = k$:

$$(1-k)x^2 - 6x + 16 + 4k = 0.$$

The discriminant condition gives $4k^2 + 9k - 4 \geq 0$, hence $k \leq -\frac{7}{2}$ or $k \geq \frac{1}{2}$. The range is $\{f(x) : f(x) \in \mathbb{R}, f(x) \leq -\frac{7}{2}\} \cup \{f(x) : f(x) \in \mathbb{R}, f(x) \geq \frac{1}{2}\}$.

8c) The graph has vertical asymptotes $x = -2$ and $x = 2$, horizontal asymptote $y = 1$, and the stationary values $-\frac{7}{2}$ and $\frac{1}{2}$.



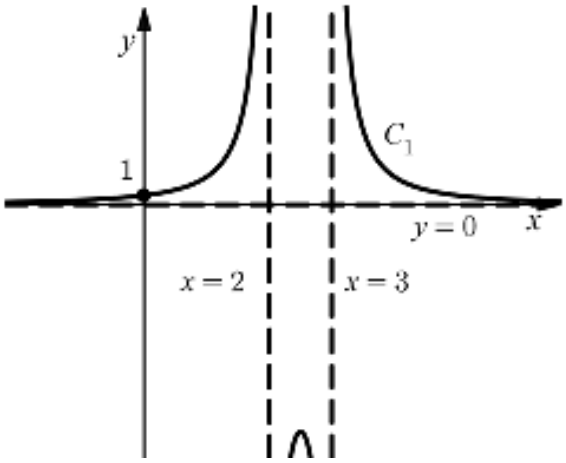
8d) For $f(x) = -4$, the solutions are $x = 0$ and $x = \frac{6}{5}$. For $f(x) = 1$, $x = \frac{10}{3}$. Testing intervals gives $0 \leq x \leq \frac{6}{5}$ or $x > \frac{10}{3}$.

9a) Since $x^2 + 1 > 0$, $f(x) \geq 0$ requires $x^2 + ax + 4 \geq 0$ for every real x. Its discriminant must be non-positive: $a^2 - 16 \leq 0$, so $a^2 \leq 16$.

9b) For $a = 4$, $f(x) = \dfrac{(x+2)^2}{x^2+1}$. Setting $f(x) = k$ and requiring real intersections gives $0 \leq k \leq 5$. The range is $\{f(x) : f(x) \in \mathbb{R}, 0 \leq f(x) \leq 5\}$.

9c) Solving $f(x) = 2$ gives $x = 2 \pm \sqrt{6}$. Since $f(x) \leq 2$ outside the interval between the roots, the solution is $x \leq 2 - \sqrt{6}$ or $x \geq 2 + \sqrt{6}$.

10a) C_1 has vertical asymptotes $x = 2, 3$, horizontal asymptote $y = 0$, and is positive for $x < 2$ or $x > 3$.



10b) At an intersection, $y = \frac{6}{x^2-5x+6}$ and $y^2 = \frac{6}{x^2-5x+6}$, so $y^2 = y$. The case $y = 0$ is impossible; $y = 1$ gives $x^2 - 5x = 0$. The points are $(0, 1)$ and $(5, 1)$.

10c) Put $u = \dfrac{6}{x^2 - 5x + 6}$. The square root requires $u \geq 0$, so $x < 2$ or $x > 3$. For $u \geq 0$, $u > \sqrt{u}$ exactly when $u > 1$, which gives $x^2 - 5x < 0$. Combining the conditions gives $0 < x < 2$ or $3 < x < 5$.

11a) $f(x) = x + \dfrac{1}{x}$. The asymptotes are $x = 0$ and $y = x$.

11b) Setting $f(x) = k$ gives $x^2 - kx + 1 = 0$. A real solution requires $k^2 - 4 \geq 0$, so the range is $\{f(x) : f(x) \in \mathbb{R}, f(x) \leq -2\} \cup \{f(x) : f(x) \in \mathbb{R}, f(x) \geq 2\}$.

11c) $f(x) > x$ is equivalent to $1/x > 0$, so $x > 0$.

12a)

Polynomial long division:

$$x^2 - 3 - (x)(x + 2) = -2x - 3,$$
$$-2x - 3 - (-2)(x + 2) = 1.$$

$f(x) = x - 2 + \dfrac{1}{x + 2}$. The asymptotes are $x = -2$ and $y = x - 2$.

12b) Setting $f(x) = k$ gives $x^2 - kx - 3 - 2k = 0$. The discriminant condition is $k^2 + 8k + 12 \geq 0$, so $k \leq -6$ or $k \geq -2$. This is the range.

12c) $f(x) \leq x - 2$ is equivalent to $\dfrac{1}{x + 2} \leq 0$, giving $x < -2$.

13a) The domain is $\{x \in \mathbb{R} : x \neq 2\}$.

13b)

Polynomial long division:

$$3x^2 - 2x + 19 - (3x)(x-2) = 4x + 19,$$
$$4x + 19 - (4)(x-2) = 27.$$

Division gives $f(x) = 3x + 4 + \dfrac{27}{x-2}$. The asymptotes are $x = 2$ and $y = 3x + 4$.

13c) Setting $f(x) = k$ yields $3x^2 + (-2-k)x + 19 + 2k = 0$. The discriminant condition gives $k^2 - 20k - 224 \geq 0$, hence $k \leq -8$ or $k \geq 28$. This is the range.

14a) The domain is $\{x \in \mathbb{R} : x \neq -1\}$.

14b)

Polynomial long division:

$$2x^2 + 5x - 3 - (2x)(x+1) = 3x - 3,$$
$$3x - 3 - (3)(x+1) = -6.$$

$f(x) = 2x + 3 - \dfrac{6}{x+1}$. The asymptotes are $x = -1$ and $y = 2x + 3$.

14c) Setting $f(x) = k$ gives $2x^2 + (5-k)x - 3 - k = 0$. Its discriminant is $(k-1)^2 + 48 > 0$ for every real k, so the range is all real numbers.

15a) The domain is $\{x \in \mathbb{R} : x \neq -4\}$.

15b)

Polynomial long division:

$$ax + x^2 + 1 - (x)(x+4) = ax - 4x + 1,$$
$$ax - 4x + 1 - (a-4)(x+4) = 17 - 4a.$$

Division gives $f(x) = x + (a-4) + \dfrac{17-4a}{x+4}$. For the oblique asymptote to be $y = x$, $a = 4$.

15c) For $a = 4$, $f(x) = x + \dfrac{1}{x+4}$. Setting $f(x) = k$ gives $x^2 + (4-k)x + 1 - 4k = 0$. The discriminant condition is $(k+2)(k+6) \geq 0$, so the range is $k \leq -6$ or $k \geq -2$. The source solution labels the vertical asymptote $x = -1$ in its division line; the question denominator gives $x = -4$.

16a)

Polynomial long division:

$$ax + x^2 + 4 - (x)(x-3) = ax + 3x + 4,$$
$$ax + 3x + 4 - (a+3)(x-3) = 3a + 13.$$

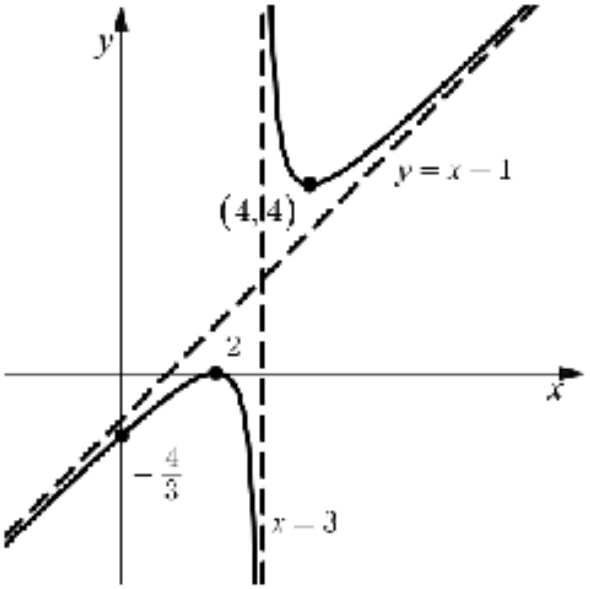
Division gives $f(x) = x + (a+3) + \dfrac{15+3a}{x-3}$. The oblique asymptote is $y = x + a + 3$.

16b) Passing through $(0, -1)$ gives $a + 3 = -1$, so $a = -4$.

16c) The domain is $\{x \in \mathbb{R} : x \neq 3\}$.

16d) For $a = -4$, $f(x) = \dfrac{x^2 - 4x + 4}{x - 3}$. Setting $f(x) = k$ gives $x^2 + (-4 - k)x + 4 + 3k = 0$. The discriminant condition is $k^2 - 4k \geq 0$, so $k \leq 0$ or $k \geq 4$. This is the range.

16e) The curve has vertical asymptote $x = 3$, oblique asymptote $y = x - 1$, and stationary values 0 and 4.



17a)

Polynomial long division:

$$2x^2 + x - 6 - (2x)(x - 2) = 5x - 6,$$
$$5x - 6 - (5)(x - 2) = 4.$$

$f(x) = 2x + 5 + \dfrac{4}{x - 2}$. The asymptotes are $x = 2$ and $y = 2x + 5$.

17b) Setting $f(x) = k$ gives $2x^2 + (1 - k)x + 2k - 6 = 0$. The discriminant condition gives $k^2 - 18k + 49 \geq 0$, so $k \leq 9 - 4\sqrt{2}$ or $k \geq 9 + 4\sqrt{2}$. This is the range.

17c) The inequality is $f(x) > 2x + 6$. Equating the curve with the boundary line gives $x = 6$. Comparing the curve with the line on its domain gives $2 < x < 6$.

18a)

Polynomial long division:

$$ax + x^2 - 1 - (x)(x + 1) = ax - x - 1,$$
$$ax - x - 1 - (a - 1)(x + 1) = -a.$$

$f(x) = x + a - 1 - \dfrac{a}{x + 1}$. The asymptotes are $x = -1$ and $y = x + a - 1$.

18b) $f(2) = \dfrac{3+2a}{3} = \frac{1}{3}$, so $a = -1$.

18c) For $a = -1$, $f(x) = \dfrac{x^2 - x - 1}{x+1} = x - 2 + \dfrac{1}{x+1}$. Setting $f(x) = k$ gives $x^2 + (-1-k)x - k - 1 = 0$. The discriminant condition is $(k+1)(k+5) \geq 0$, so the range is $k \leq -5$ or $k \geq -1$.

18d) Solving $f(x) = x + 1$ gives $x = -\frac{2}{3}$. Testing on each side, while excluding $x = -1$, gives $-1 < x \leq -\frac{2}{3}$.

19a)

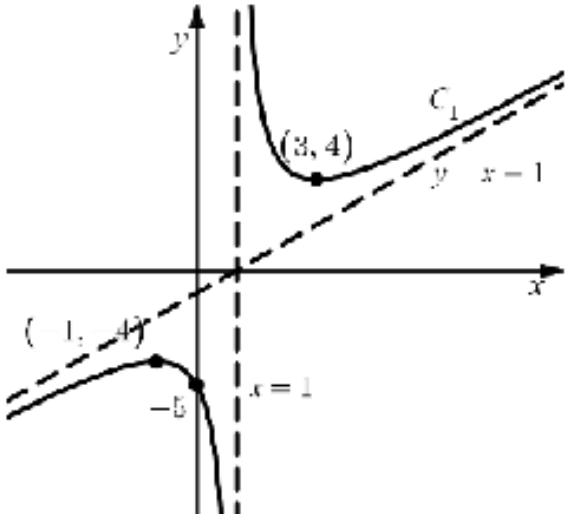
Polynomial long division:

$$x^2 - 2x + 5 - (x)(x-1) = 5 - x,$$
$$5 - x - (-1)(x-1) = 4.$$

$f(x) = x - 1 + \dfrac{4}{x-1}$. The asymptotes are $x = 1$ and $y = x - 1$.

19b) Setting $f(x) = k$ gives $x^2 + (-2-k)x + k + 5 = 0$. The discriminant condition gives $(k-4)(k+4) \geq 0$, so the range is $k \leq -4$ or $k \geq 4$.

19c) The turning points are $(-1, -4)$ and $(3, 4)$; the y-intercept is $(0, -5)$.



19d) At an intersection of $y = f(x)$ and $y = g(x) = f(x)^2$, $f(x) = f(x)^2$, so $f(x) = 0$ or $f(x) = 1$. Neither equation has a real solution: the relevant discriminants are $4 - 20 < 0$ and $9 - 24 < 0$. There are no intersections.

19e) The square root requires $f(x) \geq 0$. From the range and the graph, this occurs for $x > 1$; then $f(x) > \sqrt{f(x)}$ when $f(x) > 1$. The equality $f(x) = 1$ has no real roots, so the solution is $x > 1$.